CSA5111 - cryptography and Network security

CO1 - AT-1 :- Application Based Questions

1) a) caeser cipher - shift of 4 positions

Plain text: MEET AT NOON

M → Q , E → I → E → I ; T → X , A → E , T → X ,

N → R , O → S , O → S , N → R

Encrypted message : QIIX EX RSSR

b) Decryption process

An authorized member shifts each encrypted letter backward by 4 positions to recover the original message.

Q → M , I → E , X → T , R → N , S → O

decrypted message :- MEET AT NOON

2) a) cipher text : GSRH RH Z HVXIVG

The message is encrypted using the Atbash cipher, which replaces each letter with it's corresponding letter in reverse alphabetical order.

G → T ; S → H ; R → I ; H → S

original message: THIS IS A SECRET

b) Justification

* The ciphertext contains recognizable word patterns.
* The Atbash cipher replaces A with Z, B with Y, C with X. and so on.
* Applying the reverse-aphabet substitution produces the meaningful sentence "THIS IS A SECRET".
* The decrypted words fit the context of confidential communication.

3) a) plain text: HELLD
Key: KEY
⇒ HELLO / KEYKE

The vigenere cipher uses a keyword to encrypt a message. Repeat the keyword until it matches the plain text

A=0 B=1 .... Z=25

$C = (P + K) \bmod 26$

H ⇒ (7+10) mod 26 ⇒ R

E ⇒ (4+4) mod 26 ⇒ I

L ⇒ (11+24) mod 26 ⇒ 7

L ⇒ (11+10) mod 26 ⇒ V

O ⇒ (14+4) mod 26 ⇒ S

Encrypted message: R I J V S

b) * The keyword key is repeated to match t plaintext length.

* Each plaintext character is combined wi it's corresponding key character.

* Different key letters produce different shi

* The same plaintext letter can produce different ciphertext letters when paired wi different key letters.

4) a) Visual Inspection: Examine the image for unusual or colours, disortions or patterns

File size analysis: check whether the file s is unusually large compared with simila images.

statistical analysis: Analyze pixel values color distributions for suspicious patterns.

Stegnography detection tools: uses tools such as stegexpose to help identify potential da

Metadata Examination :- Inspect image metadata for unusual information or inconsistencies.

b) Encryption before Embedding :-

Encrypt the confendtial message using a strong encryption algorithm.

Use advanced stegnography :-

Embed the encrypted message using a suitable such as adaptive last significant bit (LSB)

5) a) cipher text : WECRLTEERDSOEEFEAOCAIVDEN

Reverse the three rail transposition by arrang the letter according to rail fence pattern.

Read the letter in the original zigzag order.

Decrypted message :

WE ARE DISCOVERED FLEE AT ONCE

b) * It only rearrange letters without changing them
* The original letter frequencies remaining unchanged.
* Attackers can try different rail arrangement to decrypt the message.